



附录：数学基础

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Outline

- **Part 1. Basic Knowledge**
 - Calculus
 - Linear Algebra
 - Probability & Statistics
- **Part 2. Optimization for ML**
 - Convex Optimization
 - Optimality Conditions and KKT Conditions
 - Gradient Descent

Notational Convention

- $[n] = \{1, \dots, n\}$
- $\mathbf{x}, \mathbf{y}, \mathbf{v}$: vectors
- A, B : matrices
- $\mathcal{X}, \mathcal{Y}, \mathcal{K}$: domain
- d : dimension
- I : identity matrix
- X, Y : random variables
- p, q : probability distributions

Calculus

- Gradient and Derivatives
- Hessian
- Chain Rule

Gradient and Derivatives (First Order)

- The gradient and derivative of a scalar function ($f : \mathbb{R} \mapsto \mathbb{R}$) is the same.
- The derivative of vector functions ($f : \mathcal{X} \subseteq \mathbb{R}^d \mapsto \mathbb{R}$) is the transpose of its gradient.

We focus on the “gradient” language (i.e., column vector).

Definition 1 (Gradient). Let $f : \mathcal{X} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable function. Let $\mathbf{x} = [x_1, \dots, x_d]^\top \in \mathcal{X}$. Then, the gradient of f at $\mathbf{x}$ is a **vector** in $\mathbb{R}^d$ denoted by $\nabla f(\mathbf{x})$ and defined by

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_d}(\mathbf{x}) \end{bmatrix}.$$

Example

Example 1. The gradient of $f(\mathbf{x}) = \|\mathbf{x}\|_2^2 \triangleq \sum_{i=1}^d x_i^2$ is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2x_1 \\ \vdots \\ 2x_d \end{bmatrix} = 2\mathbf{x}.$$

Example 2. The gradient of $f(\mathbf{x}) = -\sum_{i=1}^d x_i \ln x_i$ is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} -(\ln x_1 + 1) \\ \vdots \\ -(\ln x_d + 1) \end{bmatrix}.$$

Hessian (Second Order)

Definition 2 (Hessian). Let $f : \mathcal{X} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ be a twice differentiable function. Let $\mathbf{x} = [x_1, \dots, x_d]^\top \in \mathcal{X}$. Then, the Hessian of f at $\mathbf{x}$ is the **matrix** in $\mathbb{R}^{d \times d}$ denoted by $\nabla^2 f(\mathbf{x})$ and defined by

$$\nabla^2 f(\mathbf{x}) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) \right]_{1 \leq i, j \leq d}.$$

Example 3. The Hessian of $f(\mathbf{x}) = -\sum_{i=1}^d x_i \ln x_i$ is

$$\nabla^2 f(\mathbf{x}) = \text{diag}\left(-\frac{1}{x_1}, \dots, -\frac{1}{x_d}\right).$$

Example 4. The Hessian of $f(\mathbf{x}) = x_1^3 x_2^2 - 3x_1 x_2^3 + 1$ is

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 6x_1 x_2^2 & 6x_1^2 x_2 - 6x_2 \\ 6x_1^2 x_2 - 9x_2^2 & 2x_1^3 - 18x_1 x_2 \end{bmatrix}$$

Chain Rule

- Consider scalar functions for simplicity.

Chain Rule. For $h(x) = f(g(x))$,

- the gradient of $h(x)$ is $h'(x) = f'(g(x))g'(x)$.
- the Hessian of $h(x)$ is $h''(x) = f''(g(x))(g'(x))^2 + f'(g(x))g''(x)$.

Reference: The Matrix Cookbook

Details of **vectors, matrices, norms, determinants, etc** can be found therein.

2.4.1 First Order

$$\frac{\partial \mathbf{x}^T \mathbf{a}}{\partial \mathbf{x}} = \frac{\partial \mathbf{a}^T \mathbf{x}}{\partial \mathbf{x}} = \mathbf{a} \quad (69)$$

$$\frac{\partial \mathbf{a}^T \mathbf{X} \mathbf{b}}{\partial \mathbf{X}} = \mathbf{a} \mathbf{b}^T \quad (70)$$

$$\frac{\partial \mathbf{a}^T \mathbf{X}^T \mathbf{b}}{\partial \mathbf{X}} = \mathbf{b} \mathbf{a}^T \quad (71)$$

$$\frac{\partial \mathbf{a}^T \mathbf{X} \mathbf{a}}{\partial \mathbf{X}} = \frac{\partial \mathbf{a}^T \mathbf{X}^T \mathbf{a}}{\partial \mathbf{X}} = \mathbf{a} \mathbf{a}^T \quad (72)$$

$$\frac{\partial \mathbf{X}}{\partial X_{ij}} = \mathbf{J}^{ij} \quad (73)$$

$$\frac{\partial (\mathbf{X} \mathbf{A})_{ij}}{\partial X_{mn}} = \delta_{im} (\mathbf{A})_{nj} = (\mathbf{J}^{mn} \mathbf{A})_{ij} \quad (74)$$

$$\frac{\partial (\mathbf{X}^T \mathbf{A})_{ij}}{\partial X_{mn}} = \delta_{in} (\mathbf{A})_{mj} = (\mathbf{J}^{nm} \mathbf{A})_{ij} \quad (75)$$

<https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf>

Reference: The Matrix Cookbook

Details of **vectors, matrices, norms, determinants, etc** can be found therein.

2 Derivatives

This section is covering differentiation of a number of expressions with respect to a matrix $\mathbf{X}$. Note that it is always assumed that $\mathbf{X}$ has *no special structure*, i.e. that the elements of $\mathbf{X}$ are independent (e.g. not symmetric, Toeplitz, positive definite). See section 2.8 for differentiation of structured matrices. The basic assumptions can be written in a formula as

$$\frac{\partial X_{kl}}{\partial X_{ij}} = \delta_{ik}\delta_{lj} \quad (32)$$

that is for e.g. vector forms,

$$\left[\frac{\partial \mathbf{x}}{\partial y} \right]_i = \frac{\partial x_i}{\partial y} \quad \left[\frac{\partial \mathbf{x}}{\partial \mathbf{y}} \right]_i = \frac{\partial x}{\partial y_i} \quad \left[\frac{\partial \mathbf{x}}{\partial \mathbf{y}} \right]_{ij} = \frac{\partial x_i}{\partial y_j}$$

The following rules are general and very useful when deriving the differential of an expression ([19]):

$$\frac{\partial \mathbf{A}}{\partial \mathbf{X}} = 0 \quad (\mathbf{A} \text{ is a constant}) \quad (33)$$

$$\frac{\partial (\alpha \mathbf{X})}{\partial \mathbf{X}} = \alpha \frac{\partial \mathbf{X}}{\partial \mathbf{X}} \quad (34)$$

$$\frac{\partial (\mathbf{X} + \mathbf{Y})}{\partial \mathbf{X}} = \frac{\partial \mathbf{X}}{\partial \mathbf{X}} + \frac{\partial \mathbf{Y}}{\partial \mathbf{X}} \quad (35)$$

$$\frac{\partial (\text{Tr}(\mathbf{X}))}{\partial \mathbf{X}} = \text{Tr}(\frac{\partial \mathbf{X}}{\partial \mathbf{X}}) \quad (36)$$

$$\frac{\partial (\mathbf{X}\mathbf{Y})}{\partial \mathbf{X}} = (\frac{\partial \mathbf{X}}{\partial \mathbf{X}})\mathbf{Y} + \mathbf{X}(\frac{\partial \mathbf{Y}}{\partial \mathbf{X}}) \quad (37)$$

$$\frac{\partial (\mathbf{X} \circ \mathbf{Y})}{\partial \mathbf{X}} = (\frac{\partial \mathbf{X}}{\partial \mathbf{X}}) \circ \mathbf{Y} + \mathbf{X} \circ (\frac{\partial \mathbf{Y}}{\partial \mathbf{X}}) \quad (38)$$

$$\frac{\partial (\mathbf{X} \otimes \mathbf{Y})}{\partial \mathbf{X}} = (\frac{\partial \mathbf{X}}{\partial \mathbf{X}}) \otimes \mathbf{Y} + \mathbf{X} \otimes (\frac{\partial \mathbf{Y}}{\partial \mathbf{X}}) \quad (39)$$

$$\frac{\partial (\mathbf{X}^{-1})}{\partial \mathbf{X}} = -\mathbf{X}^{-1}(\frac{\partial \mathbf{X}}{\partial \mathbf{X}})\mathbf{X}^{-1} \quad (40)$$

$$\frac{\partial (\det(\mathbf{X}))}{\partial \mathbf{X}} = \text{Tr}(\text{adj}(\mathbf{X})\frac{\partial \mathbf{X}}{\partial \mathbf{X}}) \quad (41)$$

$$\frac{\partial (\det(\mathbf{X}))}{\partial \mathbf{X}} = \det(\mathbf{X})\text{Tr}(\mathbf{X}^{-1}\frac{\partial \mathbf{X}}{\partial \mathbf{X}}) \quad (42)$$

$$\frac{\partial (\ln(\det(\mathbf{X})))}{\partial \mathbf{X}} = \text{Tr}(\mathbf{X}^{-1}\frac{\partial \mathbf{X}}{\partial \mathbf{X}}) \quad (43)$$

$$\frac{\partial \mathbf{X}^T}{\partial \mathbf{X}} = (\frac{\partial \mathbf{X}}{\partial \mathbf{X}})^T \quad (44)$$

$$\frac{\partial \mathbf{X}^H}{\partial \mathbf{X}} = (\frac{\partial \mathbf{X}}{\partial \mathbf{X}})^H \quad (45)$$

<https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf>

Linear Algebra

- Positive (Semi-)Definite Matrix
- Rank
- Inner Product, Norm, Matrix Norm
- Matrix Decomposition

Positive (Semi-)Definite Matrix

Definition 3 (Positive Definite, PD). A matrix $A \in \mathbb{R}^{d \times d}$ is positive definite, if for all $\mathbf{x} \neq \mathbf{0}$, $\mathbf{x}^\top A \mathbf{x} > 0$, usually denoted as $A \succ 0$.

Definition 4 (Positive Semi-Definite, PSD). A matrix $A \in \mathbb{R}^{d \times d}$ is positive semi-definite, if for all $\mathbf{x} \in \mathbb{R}^d$, $\mathbf{x}^\top A \mathbf{x} \geq 0$, usually denoted as $A \succeq 0$.

Rank

- **Rank:** dimension of the vector space spanned by its columns, or the maximal number of linearly independent columns.

Example 5.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix} \xrightarrow{2R_1 + R_2 \rightarrow R_2} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 3 & 5 & 0 \end{bmatrix} \xrightarrow{-3R_1 + R_3 \rightarrow R_3} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & -1 & -3 \end{bmatrix} \\ \xrightarrow{R_2 + R_3 \rightarrow R_3} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{-2R_2 + R_1 \rightarrow R_1} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

$\Rightarrow$ The rank of matrix A is 2.

Inner Product

- Vector Space: consider $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, then

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^d x_i y_i$$

- Matrix Space: consider $A, B \in \mathbb{R}^{m \times n}$, then

$$\langle A, B \rangle = \text{Tr}(A^\top B) = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij}$$

Norm

- Typically used vector norms.

- ℓ_1 -norm:

$$\|\mathbf{x}\|_1 = |x_1| + \cdots + |x_d|$$

- ℓ_2 -norm:

$$\|\mathbf{x}\|_2 = (\mathbf{x}^\top \mathbf{x})^{1/2} = \sqrt{x_1^2 + \cdots + x_d^2} \quad \text{or called}$$

Euclidean norm

- ℓ_∞ -norm:

$$\|\mathbf{x}\|_\infty = \max \{|x_1|, \dots, |x_d|\}$$

Norm

- Typically used vector norms.
 - General ℓ_p -norm:

$$\|\mathbf{x}\|_p = (|x_1|^p + \cdots + |x_d|^p)^{1/p}$$

- Quadratic norm:

$$\|\mathbf{x}\|_A = \sqrt{\mathbf{x}^\top A \mathbf{x}}, \text{ where } A \text{ is positive semi-definite.}$$

Dual Norm

Let $\|\cdot\|$ be a vector norm on $\mathbb{R}^d$.

The associated dual norm $\|\cdot\|_*$ is defined as

$$\|\mathbf{y}\|_* = \sup \{ \mathbf{y}^\top \mathbf{x} \mid \|\mathbf{x}\| \leq 1 \}.$$

Proposition 1. The dual of ℓ_p -norm is the ℓ_q -norm with $\frac{1}{p} + \frac{1}{q} = 1$.

e.g., the dual of ℓ_2 -norm is still ℓ_2 -norm,

the dual of ℓ_1 -norm is ℓ_∞ -norm.

Proposition 2. Hölder's inequality: $\langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|_*$.

Norm Relationship

Qualitative:

Lemma 1 (Mathematical Equivalence of Norms). *Suppose that $\|\cdot\|_a$ and $\|\cdot\|_b$ are norms on $\mathbb{R}^d$, there exist positive “constants” α and β , for all $\mathbf{x} \in \mathbb{R}^d$, such that*

$$\alpha\|\mathbf{x}\|_a \leq \|\mathbf{x}\|_b \leq \beta\|\mathbf{x}\|_a.$$

Notice: constants may depend on dimension!

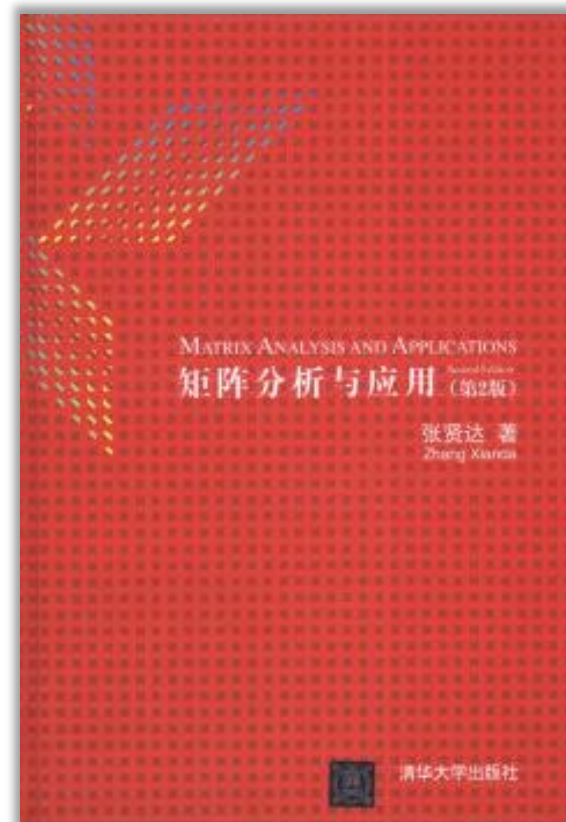
For example: for any $\mathbf{x} \in \mathbb{R}^d$, the following inequalities hold:

- $\frac{1}{d}\|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_1$
- $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \sqrt{d}\|\mathbf{x}\|_\infty$

Matrix Norm

Three different versions:

- operator norm
- entrywise norm
- Schatten norm



矩阵分析与应用. 张贤达

related pages can be found in [here](#)

Matrix Operator Norm

- Consider a matrix $A \in \mathbb{R}^{m \times n}$

We define its *operator norm* based on the *vector norm*.

Definition 5 (Matrix Operator Norm). The operator norm (or called induced norm) of a matrix $A \in \mathbb{R}^{m \times n}$ is defined by

$$\|A\|_{\text{op},p} \triangleq \max \left\{ \frac{\|A\mathbf{x}\|_p}{\|\mathbf{x}\|_p} \mid \mathbf{x} \in \mathbb{R}^d, \mathbf{x} \neq \mathbf{0} \right\}.$$

the norm in the right-hand side is defined over the *vector space*.

Matrix Operator Norm

- Consider a matrix $A \in \mathbb{R}^{m \times n}$

- ℓ_1 -norm (max-column-sum norm):

$$\|A\|_{\text{op},1} = \max_{j \in [n]} \sum_{i=1}^m |A_{ij}|$$

- ℓ_∞ -norm (max-row-sum norm):

$$\|A\|_{\text{op},\infty} = \max_{i \in [m]} \sum_{j=1}^n |A_{ij}|$$

- ℓ_2 -norm (spectral norm):

$$\|A\|_{\text{op},2} = \max_{i \in [r]} |\sigma_i|$$

where $A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$, namely, σ_i is the i -th singular value.

Matrix Entrywise Norm

- Consider a matrix $A \in \mathbb{R}^{m \times n}$

We define its *entrywise norm* by *crushing matrices as vectors*.

Definition 6 (Matrix Entrywise Norm). The entrywise norm of a matrix $A \in \mathbb{R}^{m \times n}$ is defined by

$$\|A\|_{\text{en},p} \triangleq \left(\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^p \right)^{1/p}.$$

Matrix Entrywise Norm

- Consider a matrix $A \in \mathbb{R}^{m \times n}$

- ℓ_1 -norm (sum norm):
$$\|A\|_{\text{en},1} = \sum_{i=1}^m \sum_{j=1}^n |A_{ij}|$$

- Frobenius-norm:
$$\|A\|_{\text{F}} = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2}$$

- ℓ_∞ -norm (max norm):
$$\|A\|_{\text{en},\infty} = \max_{i \in [m]} \max_{j \in [n]} |A_{ij}|$$

Eigen Value Decomposition

Let A be an $d \times d$ PSD matrix, then it can be factored as

$$A = Q\Lambda Q^\top,$$

where (a) $Q = (\mathbf{v}_1, \dots, \mathbf{v}_d) \in \mathbb{R}^{d \times d}$ is orthogonal, i.e., $Q^\top Q = I$ and $\mathbf{v}_1, \dots, \mathbf{v}_d$ are eigenvectors; and (b) $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ and $\lambda_1, \dots, \lambda_d$ are eigenvalues.

Some concerned terms can be expressed by eigenvalues:

- $A = \sum_{i=1}^d \lambda_i \mathbf{v}_i \mathbf{v}_i^\top$
- $\|A\|_{\text{op},2} = \max_{i \in [d]} |\lambda_i|$
- $\det(A) = \prod_{i=1}^d \lambda_i$
- $\text{Tr}(A) = \sum_{i=1}^d \lambda_i$
- $\|A\|_F = \sqrt{\sum_{i=1}^d \lambda_i^2}$

Singular Value Decomposition

Suppose $A \in \mathbb{R}^{m \times n}$ has rank r , then it can be factored as

$$A = U\Sigma V^\top,$$

where $U = (\mathbf{u}_1, \dots, \mathbf{u}_r) \in \mathbb{R}^{m \times r}$ satisfies $U^\top U = I$, $V = (\mathbf{v}_1, \dots, \mathbf{v}_r) \in \mathbb{R}^{n \times r}$ satisfies $V^\top V = I$; and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$ and $\sigma_1, \dots, \sigma_r$ are singular values.

Some concerned terms can be expressed by singular values:

$$- A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$$

$$- \|A\|_{\text{op},2} = \max_{i \in [r]} |\sigma_i| \quad - \|A\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2}$$

Schatten Norm

- Consider a matrix $A \in \mathbb{R}^{m \times n}$

We define its *Schatten norm* using the *singular values*.

Definition 7 (Matrix Schatten Norm). The Schatten norm of a matrix $A \in \mathbb{R}^{m \times n}$ with rank r is defined by

$$\|A\|_{\text{Sc},p} \triangleq \begin{cases} \left(\sum_{i=1}^r \sigma_i^p \right)^{1/p}, & \text{for } 1 \leq p < \infty \\ \max_{i \in [r]} |\sigma_i|, & \text{for } p = \infty \end{cases},$$

where $\sigma_1, \dots, \sigma_r$ are the singular values of A .

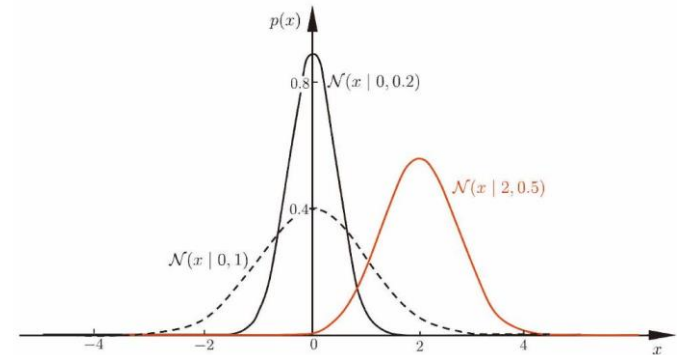
Probability and Statistics

- Typical Distributions
- Expectation and Variance
- Conditional Expectation
- Concentration Inequalities

Typical Distributions

- Gaussian Distribution

- Also called normal distribution



- d -dimensional vector $\mathbf{x}$, the parameters of the multivariate Gaussian distribution are the d -dimensional mean vector $\boldsymbol{\mu}$ and the symmetric positive definite covariance matrix $\boldsymbol{\Sigma}$ of $d \times d$. Denoted as $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$p(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^d \det(\boldsymbol{\Sigma})}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

Expectation and Variance

Expectation

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \Pr[X = x]$$

Linearity of expectation: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.

Variance

$$\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

$$\text{- Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\text{- Var}[aX] = a^2 \text{Var}[X]$$

$$\begin{aligned} \text{- Var}[X + Y] &= \text{Var}[X] + \text{Var}[Y] \\ &\quad + 2 \text{Cov}(X, Y) \end{aligned}$$

Conditional Expectation

Conditional Expectation

$$\mathbb{E}[X|Y = y] = \sum_{x \in \mathcal{X}} x \Pr[X = x|Y = y]$$

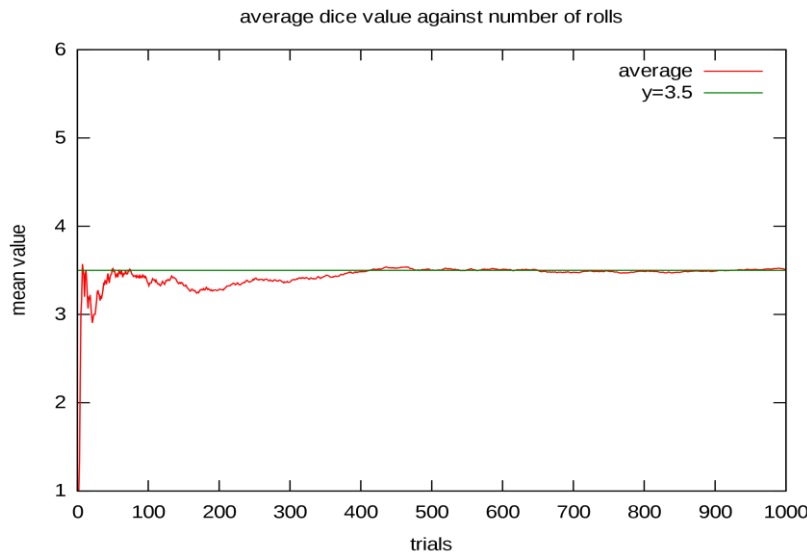
Theorem 1 (Double Expectation Theorem). *Let X, Y be arbitrary random variables. Suppose $\mathbb{E}[X], \mathbb{E}[Y], \mathbb{E}[X|Y], \mathbb{E}[Y|X]$ all exist, then it holds that*

$$\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}_X[X|Y]], \quad \mathbb{E}[Y] = \mathbb{E}_X[\mathbb{E}_Y[Y|X]].$$

$\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}_X[X|Y]]$ means that to measure the expectation of X , we can first measure the expectation of X *given the information of Y* , then measure the expectation of Y .

Law of Large Numbers

A fair, six-sided die



The average from a large number of trials should be close to the expected value and tends to become closer as more trials are performed.

Concentration Inequalities

Theorem 2 (Markov's Inequality). *Let X be a non-negative random variable with $\mathbb{E}[X] < \infty$, then for all $\epsilon > 0$,*

$$\Pr[X \geq \epsilon \mathbb{E}[X]] \leq \frac{1}{\epsilon}.$$

Theorem 3 (Chebyshev's Inequality). *Let X be a non-negative random variable with $\mathbb{E}[X], \text{Var}[X] < \infty$, then for all $\epsilon > 0$,*

$$\Pr[|X - \mathbb{E}[X]| \geq \epsilon] \leq \frac{\text{Var}[X]}{\epsilon^2}.$$

Chebyshev's inequality can be immediately obtained from Markov's inequality.

Concentration Inequalities

Theorem 4 (Hoeffding's Inequality). Let $X_1, \dots, X_m$ be *independent* random variables with X_i taking values in $[a_i, b_i]$ for all $i \in [m]$. Then, for any $\epsilon > 0$, the following inequalities hold for $S_m = \sum_{i=1}^m X_i$,

$$\Pr [S_m - \mathbb{E} [S_m] \geq \epsilon] \leq e^{-2\epsilon^2 / \sum_{i=1}^m (b_i - a_i)^2},$$
$$\Pr [S_m - \mathbb{E} [S_m] \leq -\epsilon] \leq e^{-2\epsilon^2 / \sum_{i=1}^m (b_i - a_i)^2}.$$

Part 2. Optimization for ML

- Supervised Learning
- Empirical Risk Minimization
- Convex Optimization

Learning by Optimization

The fundamental goal of (supervised) learning:

Risk Minimization (RM),

$$\min_{h \in \mathcal{H}} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [f(h(\mathbf{x}), y)],$$

where

- h denotes the hypothesis (model) from the hypothesis space $\mathcal{H}$.
- $(\mathbf{x}, y)$ is an instance chosen from a unknown distribution $\mathcal{D}$.
- $f(h(\mathbf{x}), y)$ denotes the loss of using hypothesis h on the instance $(\mathbf{x}, y)$.

Empirical Risk Minimization

The data distribution $\mathcal{D}$ is unavailable to the learner, so the risk is not computable.

In practice, the learner instead tries to optimize the empirical risk, which is called *empirical risk minimization (ERM)*:

$$\min_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m f(h(\mathbf{x}_i), y_i) .$$

ERM approximates RM:

All instances are **i.i.d.** sampled from the same distribution.

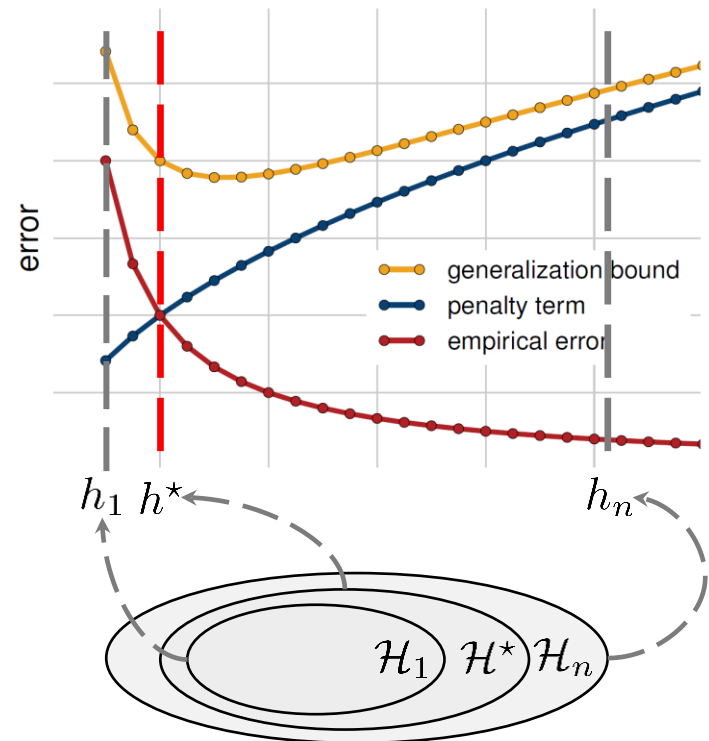
IID assumption: Independent and Identically Distributed random variables

Structural ERM

In practice, we often explicitly control the complexity of the learner by adding a regularization in optimization, i.e.,

$$\min_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m f(h(\mathbf{x}_i), y_i) + \lambda \mathcal{R}(h).$$

This is called *Structural ERM*.



Example

Consider the following binary classification task with

(i) linear hypothesis $h(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$;

(ii) $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \{-1, +1\}$ for all $i \in [m]$.

Example 6. Taking $f(h(\mathbf{x}_i), y_i) = \max\{0, 1 - y_i \mathbf{w}^\top \mathbf{x}_i\}$ (hinge loss) and $\mathcal{R}(h) = \|\mathbf{w}\|_2^2$ forms the optimization objective in *Support Vector Machine (SVM)*:

$$\min_{\mathbf{w} \in \mathbb{R}^d} \sum_{i=1}^n \max\{0, 1 - y_i \mathbf{w}^\top \mathbf{x}_i\} + \lambda \|\mathbf{w}\|_2^2.$$

Example

Consider the following binary classification task with

(i) linear hypothesis $h(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$;

(ii) $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \{-1, +1\}$ for all $i \in [m]$.

Example 7. Taking $f(h(\mathbf{x}_i), y_i) = \log(1 + \exp(-y_i \mathbf{w}^\top \mathbf{x}_i))$ and $\mathcal{R}(h) = \|\mathbf{w}\|_2^2$ forms the optimization objective in *Logistic Regression (LR)*:

$$\min_{\mathbf{w} \in \mathbb{R}^d} \sum_{i=1}^n \log(1 + \exp(-y_i \mathbf{w}^\top \mathbf{x}_i)) + \lambda \|\mathbf{w}\|_2^2.$$

Optimization

- The basic form and terminologies:

$$\begin{array}{ll}\min & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \mathcal{X}\end{array}$$

- optimization variable $\mathbf{x} \in \mathbb{R}^d$
- objective function: $f : \mathbb{R}^d \mapsto \mathbb{R}$
- feasible domain: $\mathcal{X} \subseteq \mathbb{R}^d$

Unconstrained Optimization

- The optimization variable is feasible over the whole $\mathbb{R}^d$ -space.

$$\begin{array}{ll}\min & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \mathbb{R}^d\end{array}$$

- It is one of *the most basic* forms of mathematical optimization and serves as the foundations.

--- “any optimization problem can be regarded as an unconstrained one”

$$\begin{array}{ll}\min & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \mathcal{X}\end{array} \quad \Longrightarrow \quad \begin{array}{ll}\min & h(\mathbf{x}) \triangleq f(\mathbf{x}) + \delta_{\mathcal{X}}(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \mathbb{R}^d\end{array}$$

barrier/indicator function

$$\delta_{\mathcal{X}}(\mathbf{x}) = \begin{cases} 0, & \mathbf{x} \in \mathcal{X}, \\ \infty, & \mathbf{x} \notin \mathcal{X}. \end{cases}$$

Convex Optimization

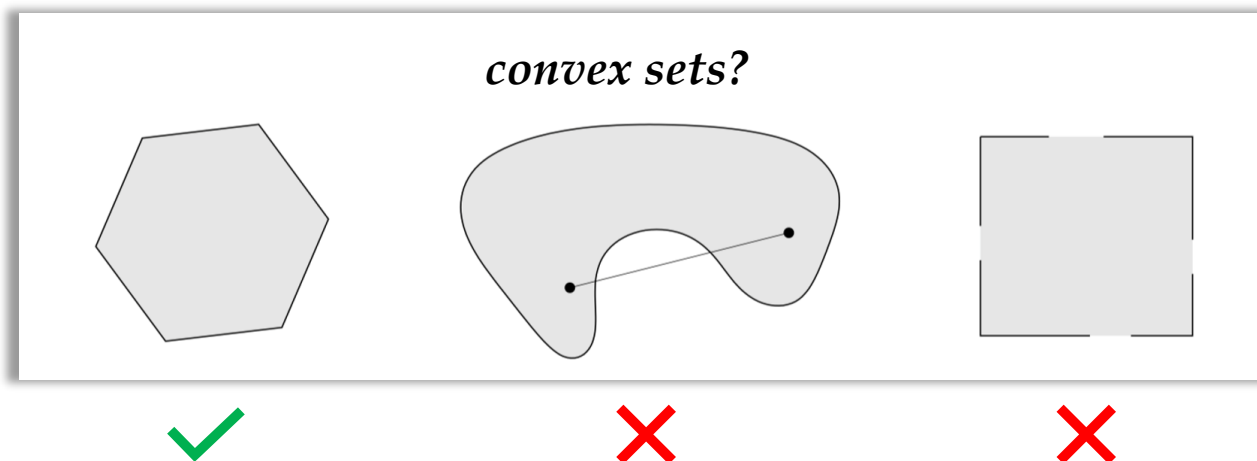
Convex optimization has a mature theory and fruitful algorithms, serving as the foundation for solving many ML problems.

- What is a convex set?
- What is a convex function?
- How to minimize?

Convex Set

Definition 1 (Convex Set). A set $\mathcal{X}$ is convex if for any $\mathbf{x}, \mathbf{y} \in \mathcal{X}$, all the points on the line segment connecting $\mathbf{x}$ and $\mathbf{y}$ also belong to $\mathcal{X}$, i.e.,

$$\forall \alpha \in [0, 1], \alpha \mathbf{x} + (1 - \alpha) \mathbf{y} \in \mathcal{X}.$$



Examples

- A line segment is convex.
- A ray, which has the form $\{\mathbf{x}_0 + \theta \mathbf{v} \mid \theta \geq 0\}$, where $\mathbf{v} \neq \mathbf{0}$, is convex.
- Any subspace is convex.

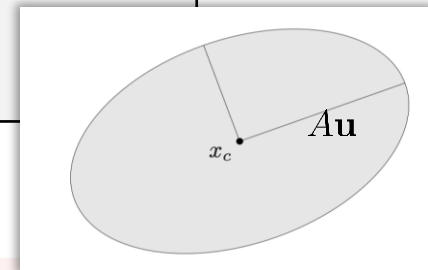
Definition 2 (Ball). A (Euclidean) ball (or just ball) in $\mathbb{R}^d$ has the form

$$\mathbb{B}(\mathbf{x}_c, r) = \{\mathbf{x}_c + \mathbf{r}\mathbf{u} \mid \|\mathbf{u}\|_2 \leq 1\}.$$

Definition 3 (Ellipsoids). An ellipsoid in $\mathbb{R}^d$ has the form

$$\mathcal{E}(\mathbf{x}_c, A) = \{\mathbf{x}_c + A\mathbf{u} \mid \|\mathbf{u}\|_2 \leq 1\},$$

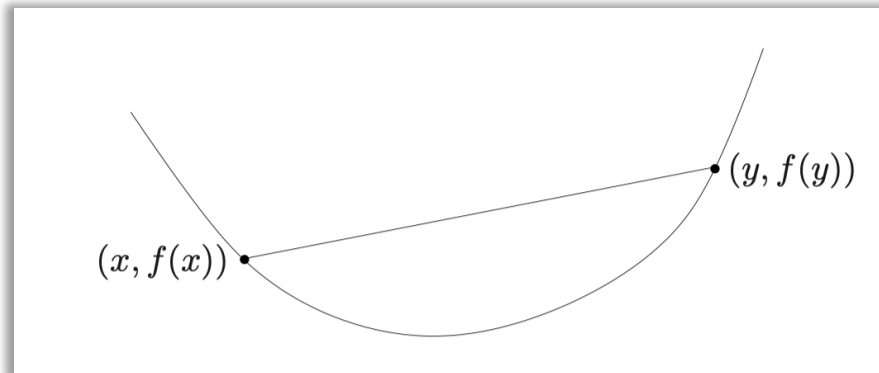
where A is assumed to be symmetric and positive definite.



Convex Function

Definition 4 (Convex Function). A function $f : \mathcal{X} \mapsto \mathbb{R}$ is convex if for any $\mathbf{x}, \mathbf{y} \in \mathcal{X}$,

$$\forall \alpha \in [0, 1], \quad f((1 - \alpha)\mathbf{x} + \alpha\mathbf{y}) \leq (1 - \alpha)f(\mathbf{x}) + \alpha f(\mathbf{y}).$$



a convex function

Convex/Concave Function

Definition 5 (Convex Function). A function $f : \mathcal{X} \mapsto \mathbb{R}$ is **convex** if for any $\mathbf{x}, \mathbf{y} \in \mathcal{X}$,

$$\forall \alpha \in [0, 1], \quad f((1 - \alpha)\mathbf{x} + \alpha\mathbf{y}) \leq (1 - \alpha)f(\mathbf{x}) + \alpha f(\mathbf{y}).$$

Definition 6 (Concave Function). A function $f : \mathcal{X} \mapsto \mathbb{R}$ is **concave** if for any $\mathbf{x}, \mathbf{y} \in \mathcal{X}$,

$$\forall \alpha \in [0, 1], \quad f((1 - \alpha)\mathbf{x} + \alpha\mathbf{y}) \geq (1 - \alpha)f(\mathbf{x}) + \alpha f(\mathbf{y}).$$

- Both definitions have already assumed a **convex** feasible domain.
- We focus on the “**convex**” language, since the negative of concave functions are convex.

Convex Function

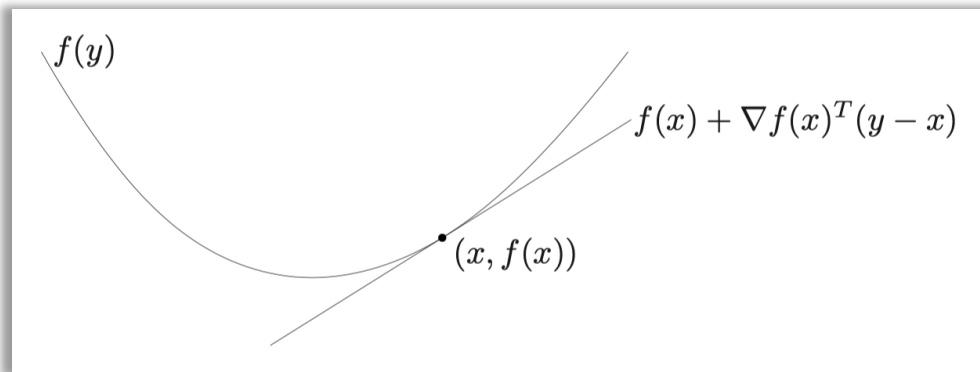
Examples on $\mathbb{R}$:

- Exponential: e^{ax} , where $a \in \mathbb{R}$.
- Powers: x^a , where $a \geq 1$ or $a \leq 0$.
- Powers of absolute value: $|x|^p$, where $p \geq 1$.
- Negative logarithm: $-\log x$.
- Negative entropy: $x \log x$.

Examples on $\mathbb{R}^d$:

- norm: $f(\mathbf{x}) = \|\mathbf{x}\|$.
- maximum: $f(\mathbf{x}) = \max \{x_1, \dots, x_n\}$.
- Log-sum-exp: $f(\mathbf{x}) = \log (e^{x_1} + \dots + e^{x_n})$.

Convex Function



If f is convex and differentiable, then $f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \leq f(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in \text{dom } f$.

the first-order Taylor approximation of f near $\mathbf{x}$

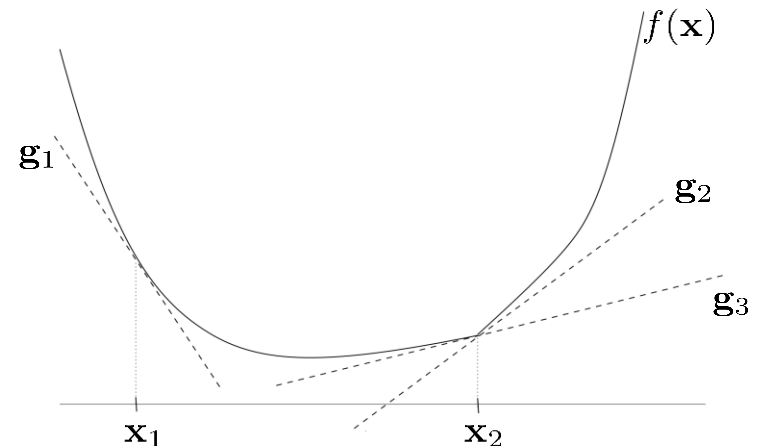
A commonly used equivalent form for analysis: $f(\mathbf{x}) - f(\mathbf{y}) \leq \langle \nabla f(\mathbf{x}), \mathbf{x} - \mathbf{y} \rangle$.

Convex Opt: (Sub-)Gradient

Definition 7 (Subgradient). Let $f : \mathcal{X} \mapsto \mathbb{R}$ be a proper function and let $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d$. A vector $\mathbf{g} \in \mathbb{R}^d$ is called a *subgradient* of f at $\mathbf{x}$ if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \mathbf{g}, \mathbf{y} - \mathbf{x} \rangle, \text{ for all } \mathbf{y} \in \mathbb{R}^d.$$

Intuition: subgradient $\mathbf{g} \in \partial f(\mathbf{x})$ can be any variable that makes the line $f(\mathbf{x}) + \langle \mathbf{g}, \mathbf{y} - \mathbf{x} \rangle$ below the curve f .



How to Compute Subgradient

- General principle: unfortunately, hard to give :(
- Ad-hoc calculations: sometimes possible, but mostly tricky
- **Good news:** easy for *convex and differential* functions.

Theorem 1. Let $f : \mathcal{X} \mapsto \mathbb{R}$ be a proper and convex function and assume $\mathcal{X}$ is convex.

1. If f is differentiable at $\mathbf{x}$, then $\partial f(\mathbf{x}) = \{\nabla f(\mathbf{x})\}$.
2. Conversely, if f has a unique subgradient, then it is differentiable at $\mathbf{x}$ and $\partial f(\mathbf{x}) = \{\nabla f(\mathbf{x})\}$.

Example 6. The subdifferential of ℓ_2 -norm $f(\mathbf{x}) = \|\mathbf{x}\|_2$ is

$$\partial f(\mathbf{x}) = \begin{cases} \left\{ \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \right\}, & \mathbf{x} \neq \mathbf{0} \text{ (gradient of norm)} \\ \{\mathbf{g} \mid \|\mathbf{g}\|_2 \leq 1\}, & \mathbf{x} = \mathbf{0} \end{cases}$$

Why Convexity?

- **Local to Global Phenomenon**

For convex (and differentiable) functions, *gradient is highly informative*.

$$\nabla f(\mathbf{x}) \in \partial f(\mathbf{x})$$

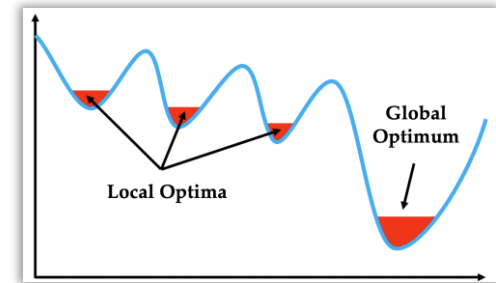
- **Local:** the gradient $\nabla f(\mathbf{x})$ contains a priori only *local* information about the function f around $\mathbf{x}$;
- **Global:** the subdifferential $\partial f(\mathbf{x})$ gives a global information in the form of a linear lower bound on the *entire* function.

Why Convexity?

- **Local to Global Phenomenon**

For convex (unconstrained) optimization, *local minima are global minima*.

Theorem 2. Let f be convex. If $\mathbf{x}$ is a local minimum of f then $\mathbf{x}$ is a global minimum of f .



A simple proof:

Assume that $\mathbf{x}$ is local minimum of f . Then for γ small enough, for any $\mathbf{y}$,

$$\underbrace{f(\mathbf{x})}_{\text{(local minima)}} \leq f((1 - \gamma)\mathbf{x} + \gamma\mathbf{y}) \leq (1 - \gamma)f(\mathbf{x}) + \gamma f(\mathbf{y}),$$

which implies $f(\mathbf{x}) \leq f(\mathbf{y})$ and thus $\mathbf{x}$ is a global minimum of f .

Optimality Condition

- Fermat's Optimality Condition
- First-order Optimality Condition
- Lagrange Multiplier and KKT Conditions

Fermat's Optimality Condition

- *Unconstrained* case

Theorem 3 (Fermat's Optimality Condition). *Let $f : \mathbb{R}^d \rightarrow (-\infty, \infty]$ be a proper convex function. Then*

$$\mathbf{x}^* \in \operatorname{argmin}\{f(\mathbf{x}) \mid \mathbf{x} \in \mathbb{R}^d\}$$

if and only if $\mathbf{0} \in \partial f(\mathbf{x}^)$.*

A simple proof:

Combining $f(\mathbf{x}) \geq f(\mathbf{x}^*)$
 $f(\mathbf{x}) \geq f(\mathbf{x}^*) + \langle \mathbf{g}, \mathbf{x} - \mathbf{x}^* \rangle, \mathbf{g} \in \partial f(\mathbf{x}^*)$ finishes the proof.

First-order Optimality Condition

- *Constrained* Case

Theorem 4 (First-order Optimality Condition). *Let f be convex and $\mathcal{X}$ a closed convex set on which f is differentiable. Then $\mathbf{x}^* \in \arg \min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$ if and only if there exists $\mathbf{g} \in \partial f(\mathbf{x}^*)$ such that*

$$\langle \mathbf{g}, \mathbf{x} - \mathbf{x}^* \rangle \geq 0, \forall \mathbf{x} \in \mathcal{X}.$$

A simple proof: derived from the *Fermat's optimality condition*.

$\Rightarrow$ deploying the Fermat's optimality condition on the unconstrained "surrogate" objective

$$h(\mathbf{x}) \triangleq f(\mathbf{x}) + \delta_{\mathcal{X}}(\mathbf{x})$$

Lagrange Multipliers

- For the optimization problem hard to solve.
- Standard form

$$\begin{array}{ll}\min_x & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0 \quad j = 1, \dots, n\end{array}$$

- feasible domain $\mathcal{X} = \{x \mid f_i(x) \leq 0, i \in [m] \text{ and } g_j(x) = 0, j \in [n]\}$, and $\mathcal{X} \neq \emptyset$
- optimal value is denoted by p^*
- we do **not** assume the problem to be convex

Lagrange Multipliers

- Define the Lagrangian function as

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^n \nu_j h_j(x)$$

- λ_i : the Lagrange multiplier associated with the i -th inequality constraint $f_i(x) \leq 0$
note that we require $\lambda_i \geq 0$ for all $i = 1, \dots, m$
- ν_j : the Lagrange multiplier associated with the j -th equality constraint $h_j(x) = 0$
- Vectors λ and ν : **dual variables** for Lagrange multiplier vectors

Lagrange Multipliers

- Define the maximum function

$$\begin{aligned} g(\lambda, \nu) &= \inf_{x \in \mathcal{X}} L(x, \lambda, \nu) \\ &= \inf_{x \in \mathcal{X}} \left(f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^n \nu_j h_j(x) \right) \end{aligned}$$

- $g(\lambda, \nu)$ is **concave** no matter the primal problem is convex or not
- this optimization for $g(\lambda, \nu)$ is **unconstrained**
- naturally serving as a **lower bound** for optimal value p^*

$$g(\lambda, \nu) \leq p^* \text{ holds for any } \lambda \succeq 0 \text{ and } \nu$$

A simple proof: $\forall \tilde{x} \in \mathcal{X}$, we have $L(\tilde{x}, \lambda, \nu) = f_0(\tilde{x}) + \sum_{i=1}^m \lambda_i f_i(\tilde{x}) + \sum_{i=1}^p \nu_i h_i(\tilde{x}) \leq f_0(\tilde{x})$

Why Lagrangian Function?

- Linear Approximation Interpretation

$$\begin{array}{ll}\min_x & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0 \quad j = 1, \dots, n\end{array}$$

$$\Rightarrow \min f_0(x) + \sum_{i=1}^m I_{-}(f_i(x)) + \sum_{j=1}^n I_0(h_j(x))$$

- $I_{-} : \mathbb{R} \mapsto \mathbb{R}$ is the indicator function for the nonpositive reals

$$I_{-}(u) = \begin{cases} 0 & u \leq 0, \\ \infty & u > 0. \end{cases}$$

- I_0 is the indicator function of $\{0\}$

Why Lagrangian Function?

- Linear Approximation Interpretation

$$\begin{array}{ll}\min_x & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0 \quad j = 1, \dots, n\end{array}$$

$$\Rightarrow \min f_0(x) + \sum_{i=1}^m I_{-}(f_i(x)) + \sum_{j=1}^n I_0(h_j(x))$$

$$\Rightarrow L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^n \nu_j h_j(x)$$

relaxation by replacing $I_{-}(u)$ by $\lambda_i u$, and $I_0(u)$ by $\nu_j u$

Why Lagrangian Function?

- Linear Approximation Interpretation

$$\begin{array}{ll}\min_x & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0 \quad j = 1, \dots, n\end{array}$$

$$\Rightarrow \min f_0(x) + \sum_{i=1}^m \textcolor{red}{I}_-(f_i(x)) + \sum_{j=1}^n \textcolor{blue}{I}_0(h_j(x))$$

$$\Rightarrow L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \textcolor{red}{\lambda}_i f_i(x) + \sum_{j=1}^n \textcolor{blue}{\nu}_j h_j(x)$$

relaxation by replacing $\textcolor{red}{I}_-(u)$ by $\textcolor{red}{\lambda}_i u$, and $\textcolor{blue}{I}_0(u)$ by $\textcolor{blue}{\nu}_j u$

The Lagrange Dual Problem

$$\begin{aligned} g(\lambda, \nu) &= \inf_{x \in \mathcal{X}} L(x, \lambda, \nu) \\ &= \inf_{x \in \mathcal{X}} \left(f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^n \nu_j h_j(x) \right) \end{aligned}$$

- As mentioned, this naturally serves as a **lower bound** for optimal value p^*

$$g(\lambda, \nu) \leq p^* \text{ holds for any } \lambda \succeq 0 \text{ and } \nu$$

- Lagrange Dual Problem

$$\begin{array}{ll} \max & g(\lambda, \nu) \\ \text{s.t.} & \lambda \succeq 0 \end{array}$$

Weak Duality

$$d^* \leq p^*$$

d^* denotes the optimal value of dual problem

This aims to obtain the best lower bound for estimating optimal value p^*

Strong Duality and KKT Conditions

- Strong Duality (defined as the following condition holds)

$$d^* = p^*$$

- The best bound that can be obtained from the Lagrange dual function is tight
- In general, does not hold, but **usually** hold for convex optimization

- Suppose strong duality holds

$$f_0(x^*) = g(\lambda^*, \nu^*)$$

(by strong duality)

$$= \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{j=1}^n \nu_j^* h_j(x) \right)$$

(by definition of dual function)

$$\stackrel{\text{hold with}}{\leq} f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{j=1}^n \nu_j^* h_j(x^*)$$

(by taking infimum)

$$\stackrel{\text{equality!}}{\leq} f_0(x^*)$$

(by requirements of constraints)

Strong Duality and KKT Conditions

- Strong Duality (defined as the following condition holds)

$$d^* = p^*$$

- The best bound that can be obtained from the Lagrange dual function is tight
- In general, does not hold, but **usually** hold for convex optimization

- Suppose strong duality holds

$$\begin{aligned} f_0(x^*) &= g(\lambda^*, \nu^*) && \text{(by strong duality)} \\ &= \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{j=1}^n \nu_j^* h_j(x) \right) && \text{(by definition of dual function)} \\ &= f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{j=1}^n \nu_j^* h_j(x^*) && \text{(by taking infimum)} \\ &= f_0(x^*) && \text{(by requirements of constraints)} \end{aligned}$$

hold with equality!

KKT Conditions

- Suppose strongly duality holds

$$\begin{aligned} f_0(x^*) &= g(\lambda^*, \nu^*) \\ &= \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{j=1}^n \nu_j^* h_j(x) \right) \\ &= f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{j=1}^n \nu_j^* h_j(x^*) \\ &= f_0(x^*) \end{aligned}$$

- Equality in the third line implies x^* minimizes $L(x, \lambda^*, \nu^*)$
- Equality in the last line implies $\sum_{i=1}^m \lambda_i^* f_i(x^*) = 0$

KKT Conditions

- x^* and (λ^*, ν^*) : any primal and dual optimal points with strong duality

- Equality in the third line implies x^* minimizes $L(x, \lambda^*, \nu^*)$

$$\Rightarrow \nabla L(x^*, \lambda^*, \nu^*) = 0$$

$$\Rightarrow \nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^n \nu_j^* \nabla h_j(x^*) = 0$$

- Equality in the last line implies $\sum_{i=1}^m \lambda_i^* f_i(x^*) = 0$

$$\Rightarrow \text{Other expressions} \quad \lambda_i^* > 0 \Rightarrow f_i(x^*) = 0$$

$$f_i(x^*) < 0 \Rightarrow \lambda_i^* = 0$$

This is also called “Complementary Slackness”

KKT Conditions

- x^* and (λ^*, ν^*) : any primal and dual optimal points with strong duality

$$\begin{aligned}f_i(x^*) &\leq 0, & i \in [m] \\h_j(x^*) &= 0, & j \in [n] \\\lambda_i^* &\geq 0, & i \in [m] \\\lambda_i^* f_i(x^*) &= 0, & i \in [m] \\\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^n \nu_j^* \nabla h_j(x^*) &= 0\end{aligned}$$

**Necessary
Condition**

For optimization problem with differentiable objective and constraint functions for which strong duality obtains, any pair of primal and dual optimal must satisfy KKT conditions.

KKT Conditions

- If f_i are convex, h_i are affine, $\tilde{x}, \tilde{\lambda}, \tilde{\nu}$ satisfy

$$f_i(\tilde{x}) \leq 0, \quad i \in [m]$$

$$h_j(\tilde{x}) = 0, \quad j \in [n]$$

$$\tilde{\lambda}_i^* \geq 0, \quad i \in [m]$$

$$\tilde{\lambda}_i^* f_i(\tilde{x}) = 0, \quad i \in [m]$$

$$\nabla f_0(\tilde{x}) + \sum_{i=1}^m \tilde{\lambda}_i^* \nabla f_i(\tilde{x}) + \sum_{j=1}^n \tilde{\nu}_j^* \nabla h_j(\tilde{x}) = 0$$

$\Rightarrow$ Then, $\tilde{x}$ and $\tilde{\lambda}, \tilde{\nu}$ are primal and dual optimal, with strong duality.

**Sufficient
Condition**

For any **convex** optimization problem with differentiable objective and constraint functions, any points that satisfy the KKT conditions are primal and dual optimal, and they enjoy strong duality.

KKT Conditions

- The KKT conditions play an important role in optimization.
 - In a few special cases it is possible to solve the KKT conditions.
 - More generally, many algorithms for convex optimization can be interpreted as methods for solving the KKT conditions.
- Karush–Kuhn–Tucker (KKT) Conditions



William Karush
1917-1997

Developed (necessary) conditions in 1939 in his (unpublished) MS thesis.



Harold Kuhn
1925-2014



Albert Tucker
1905-1995

Published the overall conditions in 1951.

Gradient Descent

- GD Template:

$$\mathbf{x}_{t+1} = \Pi_{\mathcal{X}} [\mathbf{x}_t - \eta_t \nabla f(\mathbf{x}_t)]$$

- $\mathbf{x}_1$ can be an arbitrary point inside the domain.
 - $\eta_t > 0$ is the potentially time-varying *step size* (or called *learning rate*).
 - Projection $\Pi_{\mathcal{X}}[\mathbf{y}] = \arg \min_{\mathbf{x} \in \mathcal{X}} \|\mathbf{x} - \mathbf{y}\|$ ensures the feasibility.
- Many variants: SGD, SGD + momentum, Adam, AdaGrad, ...

Summary

CALCULUS

Gradient and Derivatives

Hessian

Chain Rule

The Matrix Cookbook

PROBABILITY & STATISTICS

Expectation and Variance

Conditional Expectation

Concentration Inequalities

LINEAR ALGEBRA

Positive (Semi-)Definite Matrix

Rank

Inner Product

Norm

Norm

Matrix Norm

Norm Relationship

Decomposition

Eigen Value Decomposition

Singular Value Decomposition

OPTIMIZATION IN MACHINE LEARNING

Risk Minimization

Empirical Risk Minimization (ERM)

Convex Optimization

Optimality Conditions and KKT

Gradient Descent

Q & A

Thanks!